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B. Tech – Computer Science Engineering (Artificial Intelligence)

22AIE401 – Reinforcement Learning

Inverse Reinforcement Learning for Route Preference Learning

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Problem Background

Navigation applications such as Google Maps, Waze, and Apple Maps help users find suitable routes by considering factors such as travel distance, travel time, traffic, road conditions, and other route information. These systems use graph-based routing along with machine learning and other techniques to improve route recommendations.

However, people do not always choose the shortest or fastest route. A driver may choose a slightly longer route because it has better road conditions, fewer difficult turns, less traffic, or a more comfortable driving experience. These preferences can also vary from one driver to another.

With the availability of historical GPS trajectory data, we can study the routes that drivers choose instead of relying only on manually defined assumptions. This gives us an opportunity to learn the preferences behind human route choices.

Inverse Reinforcement Learning (IRL) is useful for this problem because it works in the opposite way of normal Reinforcement Learning. Instead of starting with a known reward and learning how to act, IRL observes human actions and tries to learn the hidden reward that explains those actions using historical GPS trajectories. The learned reward functions can then be used to generate routes that better align with human preferences.

Problem Statement

Develop an Inverse Reinforcement Learning-based route planning system that learns human driving preferences from historical GPS trajectories and generates realistic, human-aligned routes on OpenStreetMap road networks.

Dataset:

The proposed system will use historical GPS trajectory datasets such as GeoLife GPS Trajectories or other publicly available trajectory datasets. OpenStreetMap (OSM) will be used to construct the road network for route planning experiments.

Why Reinforcement Learning?

Route planning is a sequential decision-making problem. During a journey, a driver reaches different intersections and repeatedly decides which road to take next. Each decision affects the future part of the journey.

This can naturally be represented using a Markov Decision Process (MDP), where the current road location is the state, the possible roads are the actions, and the reward represents how desirable a particular choice is.

Traditional Reinforcement Learning assumes that the reward function is already known. In our problem, the reward that represents human driving preferences is not directly available. We can observe which roads drivers choose, but we cannot directly observe the reasons or preferences behind those choices.

Therefore, Inverse Reinforcement Learning is used to learn the hidden reward or preferences from observed human driving behavior.

Literature Review

S.No	Paper	Year	Relevance
1	Maximum Entropy Inverse Reinforcement Learning — Ziebart et al.	2008	Foundational IRL method; specifically developed and demonstrated for navigation and driving behaviour.
2	Integrating Dijkstra’s Algorithm into Deep Inverse Reinforcement Learning for Food Delivery Route Planning — Liu et al.	2020	Directly combines IRL with Dijkstra and learns delivery-driver preferences from GPS trajectories.
3	A Deep Inverse Reinforcement Learning Approach to Route Choice Modeling with Context-Dependent Rewards — Zhao & Liang	2023	Directly studies driver route choice using deep IRL and taxi GPS data.
4	Massively Scalable Inverse Reinforcement Learning in Google Maps — Barnes et al.	2024	Demonstrates IRL for route recommendation at Google Maps scale.
5	Recursive Logit-Based Meta-Inverse Reinforcement Learning for Driver-Preferred Route Planning — Zhang et al.	2024	Addresses computational efficiency and sparse GPS coverage in driver-preference IRL.
6	Trajectory Modeling via Random Utility Inverse Reinforcement Learning — Pitombeira-Neto et al.	2024	Directly models driver trajectories using IRL and accounts for observed/unobserved factors.

7	Personalized Origin–Destination Travel Time Estimation with Active Adversarial Inverse Reinforcement Learning and Transformer — Liu et al.	2025	Relevant to personalized route preference learning using adversarial IRL.
8	Modelling Route Choice in Public Transport with Deep Learning — Marra & Corman	2025	Supporting route-choice research showing how learned nonlinear utilities can model complex route preferences.

Existing System

Modern navigation systems such as Google Maps, Waze, and Apple Maps already use advanced routing methods along with information such as traffic, road conditions, travel time, and other signals.

Google has also published research on using Inverse Reinforcement Learning for Google Maps. Their work shows that IRL can be used to learn the hidden preferences of travellers from real-world travel patterns. The research also addresses the major challenge of scaling IRL to very large road networks. This shows that learning human route preferences using IRL is a practical research direction and not only a theoretical idea.

However, commercial navigation systems are proprietary. Their complete routing models, reward functions, and personalization methods are not publicly available. Therefore, it is difficult to directly study and reproduce their complete decision-making process. Our project focuses on creating a smaller and transparent research framework where the relationship between human trajectories, learned preferences, and route generation can be studied and evaluated.

Gap Analysis

Traditional routing algorithms like Dijkstra's and A* rely on static, manually engineered cost functions typically minimizing distance or travel time failing to capture the subjective, multi-criteria trade-offs that govern real-world driving behavior.

Supervised machine learning can predict trajectory patterns from historical data, but it acts primarily as an uninterpretable black box. It predicts which paths may be selected without recovering the underlying utility function that explains why drivers make specific decisions, limiting its generalizability across unseen origin-destination pairs.

While Inverse Reinforcement Learning (IRL) mathematically infers latent reward functions from demonstrations, existing implementations face significant hurdles: high computational overhead during forward-MDP solving on large road networks, high sensitivity to sparse or noisy GPS traces, and an ongoing trade-off between complex deep-IRL models and interpretable feature representations.

Despite recent advances, existing IRL-based route planning systems either rely on computationally intensive deep models, large-scale proprietary datasets, or black-box implementations that are difficult to reproduce and interpret. Therefore, a gap remains for a computationally efficient and transparent IRL framework that can learn explainable driver preferences from publicly available GPS trajectory data and OpenStreetMap road networks.

Proposed System

The proposed research will use historical GPS trajectories as examples of real human driving behaviour and an OpenStreetMap road network as the environment.

The road network will be represented as a Markov Decision Process, where drivers make a sequence of road-selection decisions while travelling from an origin to a destination.

Maximum Entropy Inverse Reinforcement Learning (MaxEnt IRL) will be used to learn the underlying reward function that best explains the observed route choices

The study will consider relevant route characteristics such as distance, travel time, road type, turns, and other available road or traffic information. The exact factors will depend on the data available for the selected study area.

Performance will be evaluated using route similarity measures, trajectory overlap, path deviation, and comparison against expert driver trajectories.

The main objective is to study whether a reward learned from real driving behaviour can produce routes that more closely reflect human route-selection patterns.

The project is therefore focused not on replacing existing navigation applications, but on understanding and demonstrating how Inverse Reinforcement Learning can be used to learn human route preferences in a transparent and reproducible way.

